

MONGO -DB CRUD OPERATION:

```
import com.mongodb.client.FindIterable;
import com.mongodb.client.MongoClient;
import com.mongodb.client.MongoClients;
import com.mongodb.client.MongoCollection;
import com.mongodb.client.MongoDatabase;
import com.mongodb.client.model.Filters;
import com.mongodb.client.model.Updates;
import org.bson.Document;

public class Test2 {

    public static void main(String[] args) {

        // Connect to MongoDB

        MongoClient mongoClient = MongoClients.create("mongodb://localhost:27017");

        // Access Database and Collection

        MongoDatabase database = mongoClient.getDatabase("CollegeDB");

        MongoCollection<Document> collection = database.getCollection("Students");

        // =====

        // CREATE (Insert 5 Students)

        // =====

        collection.insertOne(new Document("id", 101)

            .append("name", "Aarav")

            .append("department", "CSE")

            .append("cgpa", 8.5));

        collection.insertOne(new Document("id", 102)

            .append("name", "Priya")

            .append("department", "IT")

            .append("cgpa", 9.1));

        collection.insertOne(new Document("id", 103)

            .append("name", "Rahul")
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        .append("department", "ECE")
        .append("cgpa", 8.0));
collection.insertOne(new Document("id", 104)
        .append("name", "Sneha")
        .append("department", "ME")
        .append("cgpa", 8.8));

collection.insertOne(new Document("id", 105)
        .append("name", "Vikram")
        .append("department", "CIVIL")
        .append("cgpa", 7.9));
System.out.println("5 Documents Inserted Successfully");
// =====
// READ (Display All Students)
// =====
System.out.println("\nStudent Records:");
FindIterable<Document> students = collection.find();
for (Document doc : students) {
    System.out.println(doc.toJson());
}
// =====
// UPDATE
// Update CGPA of Student ID 103
// =====
collection.updateOne(
    Filters.eq("id", 103),
    Updates.set("cgpa", 9.3)
);

```

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System.out.println("\nStudent ID 103 Updated Successfully");

// =====

// READ Updated Record

// =====

Document updatedStudent =
    collection.find(Filters.eq("id", 103)).first();

System.out.println("\nUpdated Record:");

if (updatedStudent != null) {
    System.out.println(updatedStudent.toJson());
}

// =====

// DELETE

// Delete Student ID 105

// =====

collection.deleteOne(Filters.eq("id", 105));

System.out.println("\nStudent ID 105 Deleted Successfully");

// =====

// DISPLAY FINAL RECORDS

// =====

System.out.println("\nFinal Student Records:");

for (Document doc : collection.find()) {
    System.out.println(doc.toJson());
}

// Close Connection

mongoClient.close();
}

```

```

5 Documents Inserted Successfully

Student Records:
{"_id": {"$oid": "6a2024506c25d75be27f0868"}, "id": 101, "name": "Aarav", "department": "CSE", "cgpa": 8.5}
{"_id": {"$oid": "6a2024506c25d75be27f0869"}, "id": 102, "name": "Priya", "department": "IT", "cgpa": 9.1}
{"_id": {"$oid": "6a2024506c25d75be27f086a"}, "id": 103, "name": "Rahul", "department": "ECE", "cgpa": 8.0}
{"_id": {"$oid": "6a2024506c25d75be27f086b"}, "id": 104, "name": "Sneha", "department": "ME", "cgpa": 8.8}
{"_id": {"$oid": "6a2024506c25d75be27f086c"}, "id": 105, "name": "Vikram", "department": "CIVIL", "cgpa": 7.9}

Student ID 103 Updated Successfully

Updated Record:
{"_id": {"$oid": "6a2024506c25d75be27f086a"}, "id": 103, "name": "Rahul", "department": "ECE", "cgpa": 9.3}

Student ID 105 Deleted Successfully

Final Student Records:
{"_id": {"$oid": "6a2024506c25d75be27f0868"}, "id": 101, "name": "Aarav", "department": "CSE", "cgpa": 8.5}
{"_id": {"$oid": "6a2024506c25d75be27f0869"}, "id": 102, "name": "Priya", "department": "IT", "cgpa": 9.1}
{"_id": {"$oid": "6a2024506c25d75be27f086a"}, "id": 103, "name": "Rahul", "department": "ECE", "cgpa": 9.3}
{"_id": {"$oid": "6a2024506c25d75be27f086b"}, "id": 104, "name": "Sneha", "department": "ME", "cgpa": 8.8}

```

For Many:

package mongodb2;

import com.mongodb.client.FindIterable;

import com.mongodb.client.MongoClient;

import com.mongodb.client.MongoClients;

import com.mongodb.client.MongoCollection;

import com.mongodb.client.MongoDatabase;

import com.mongodb.client.model.Filters;

import com.mongodb.client.model.Updates;

import java.util.Arrays;

import org.bson.Document;

public class Test2 {

 public static void main(String[] args) {

 // Connect to MongoDB

 MongoClient mongoClient = MongoClients.create("mongodb://localhost:27017");

 // Access Database and Collection

 MongoDatabase database = mongoClient.getDatabase("CollegeDB");

 MongoCollection<Document> collection = database.getCollection("Students")

 // CREATE (Insert 5 Students)

 collection.insertMany(Arrays.asList(

 new Document("id", 101)

 .append("name", "Aarav")

 .append("department", "CSE")

```

        .append("cgpa", 8.5),
new Document("id", 102)
        .append("name", "Priya")
        .append("department", "IT")
        .append("cgpa", 9.1),
new Document("id", 103)
        .append("name", "Rahul")
        .append("department", "CSE")
        .append("cgpa", 8.0),
new Document("id", 104)
        .append("name", "Sneha")
        .append("department", "ME")
        .append("cgpa", 8.8),
new Document("id", 105)
        .append("name", "Vikram")
        .append("department", "CSE")
        .append("cgpa", 7.9)));

System.out.println("5 Documents Inserted Successfully");

// READ (Display All Students)

System.out.println("\nStudent Records:");

for (Document doc : collection.find()) {
    System.out.println(doc.toJson());
}

// UPDATE

collection.updateMany(
    Filters.eq("department", "CSE"),
    Updates.set("department", "Computer Science")
);

System.out.println("\nMultiple Documents Updated Successfully");

```

```

// READ Updated Record

System.out.println("\nRecords After Update:");

for (Document doc : collection.find()) {

    System.out.println(doc.toJson());

}

// Delete Many

collection.deleteMany(

    Filters.eq("department", "Computer Science")

);

System.out.println("\nMultiple Documents Deleted Successfully");

// DISPLAY FINAL RECORDS

System.out.println("\nFinal Student Records:");

for (Document doc : collection.find()) {

    System.out.println(doc.toJson());

}

// Close Connection

mongoClient.close();

}

```

```

Student Records:
{"_id": {"$oid": "6a2024506c25d75be27f0868"}, "id": 101, "name": "Aarav", "department": "CSE", "cgpa": 8.5}
{"_id": {"$oid": "6a2024506c25d75be27f0869"}, "id": 102, "name": "Priya", "department": "IT", "cgpa": 9.1}
{"_id": {"$oid": "6a2024506c25d75be27f086a"}, "id": 103, "name": "Rahul", "department": "ECE", "cgpa": 9.3}
{"_id": {"$oid": "6a2024506c25d75be27f086b"}, "id": 104, "name": "Sneha", "department": "ME", "cgpa": 8.8}
{"_id": {"$oid": "6a202fed045645123b6f8485"}, "id": 101, "name": "Aarav", "department": "CSE", "cgpa": 8.5}
{"_id": {"$oid": "6a202fed045645123b6f8486"}, "id": 102, "name": "Priya", "department": "IT", "cgpa": 9.1}
{"_id": {"$oid": "6a202fed045645123b6f8487"}, "id": 103, "name": "Rahul", "department": "CSE", "cgpa": 8.0}
{"_id": {"$oid": "6a202fed045645123b6f8488"}, "id": 104, "name": "Sneha", "department": "ME", "cgpa": 8.8}
{"_id": {"$oid": "6a202fed045645123b6f8489"}, "id": 105, "name": "Vikram", "department": "CSE", "cgpa": 7.9}

Multiple Documents Updated Successfully

Records After Update:
{"_id": {"$oid": "6a2024506c25d75be27f0868"}, "id": 101, "name": "Aarav", "department": "Computer Science", "cgpa": 8.5}
{"_id": {"$oid": "6a2024506c25d75be27f0869"}, "id": 102, "name": "Priya", "department": "IT", "cgpa": 9.1}
{"_id": {"$oid": "6a2024506c25d75be27f086a"}, "id": 103, "name": "Rahul", "department": "ECE", "cgpa": 9.3}
{"_id": {"$oid": "6a2024506c25d75be27f086b"}, "id": 104, "name": "Sneha", "department": "ME", "cgpa": 8.8}
{"_id": {"$oid": "6a202fed045645123b6f8485"}, "id": 101, "name": "Aarav", "department": "Computer Science", "cgpa": 8.5}
{"_id": {"$oid": "6a202fed045645123b6f8486"}, "id": 102, "name": "Priya", "department": "IT", "cgpa": 9.1}
{"_id": {"$oid": "6a202fed045645123b6f8487"}, "id": 103, "name": "Rahul", "department": "Computer Science", "cgpa": 8.0}
{"_id": {"$oid": "6a202fed045645123b6f8488"}, "id": 104, "name": "Sneha", "department": "ME", "cgpa": 8.8}
{"_id": {"$oid": "6a202fed045645123b6f8489"}, "id": 105, "name": "Vikram", "department": "Computer Science", "cgpa": 7.9}
}

```

